For Fun and Profit

Build your stack with Scapy

v2

Disclaimer

I do not pretend to know everything about the topics discussed in this workshop.

If you have a question about anything discussed in this workshop, ask.

There are no stupid or dumb questions, only the ones not asked.

nmap workshop -vvv

- tmux
- iPython
- 802.11
- TCP
- Scapy
- airpwn-ng && (pyDot11 | airtun-ng)
- Tools
- SNIPs

Primer Questions

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- Q: How many of you are red teamers?

tmux

- Used as a terminal multiplexer
- Allows you to re-ssh without nohup nonsense
- Essential for headless
- Saves real-estate
- Scapy wants shells.

IPython

- Tab it out
- ?
- from IPython import embed; embed()
- from scapy.all import *

802.11

- Radio waves are all around us
- Works off the concept of "Shared Medium"
- All we have to do, is listen
- Jurisdiction matters, know your laws

Useful 802.11 Terminology

- Authenticator: An entity at one end of a point-to-point LAN segment that facilitates authentication of the entity attached to the other end of that link. (IEEE Std 802.1X-2004)
- BSSID: Basic Service Set Identifier. Commonly known as the MAC address of a specific Access Point.
- Distribution system (DS): A system used to interconnect a set of basic service sets (BSSs) and integrated local area networks (LANs) to create an extended service set (ESS).
- ESSID: Extended Service Set Identifier. Commonly referred to as the SSID and known by the masses as the "Network Name".

Useful 802.11 Terminology

- Group Temporal Key (GTK): A random value, assigned by the group source, which is used to protect group addressed medium access control (MAC) protocol data units (MPDUs) from that source. The GTK might be derived from a group master key (GMK).
- Medium Access Control (MAC) Protocol Data Unit (MPDU): The unit of data exchanged between two peer MAC entities using the services of the physical layer (PHY).
- Pairwise Master Key (PMK): The key derived from a key generated by an Extensible Authentication Protocol (EAP) method or obtained directly from a preshared key (PSK).
- Pairwise Transient Key (PTK): A concatenation of session keys derived from the pairwise master key (PMK) or from the PMK-R1. Its components include a key confirmation key (KCK), a key encryption key (KEK), and one or more temporal keys that are used to protect information exchanged over the link.

Useful 802.11 Terminology

- Station (STA): A logical entity that is a single addressable instance of a medium access control (MAC) and physical layer (PHY) interface to the wireless medium (WM).
- Supplicant: An entity at one end of a point-to-point LAN segment that is being authenticated by an Authenticator attached to the other end of that link. (IEEE Std 802.1X-2004)

From-DS and To-DS

Table 8-2—To/From DS combinations in data frames

To DS and From DS values	Meaning
To DS = 0 From DS = 0	A data frame direct from one STA to another STA within the same IBSS, a data frame direct from one non-AP STA to another non-AP STA within the same BSS, or a data frame outside the context of a BSS, as well as all management and control frames.
To DS = 1 From DS = 0	A data frame destined for the DS or being sent by a STA associated with an AP to the Port Access Entity in that AP.
To DS = 0 From DS = 1	A data frame exiting the DS or being sent by the Port Access Entity in an AP, or a group addressed Mesh Data frame with Mesh Control field present using the three-address MAC header format.
To DS = 1 From DS = 1	A data frame using the four-address MAC header format. This standard defines procedures for using this combination of field values only in a mesh BSS.

Address Fields

From DS Set, To DS Clear:
Address 1: Destination
Address 2: BSSID
Address 3: Source

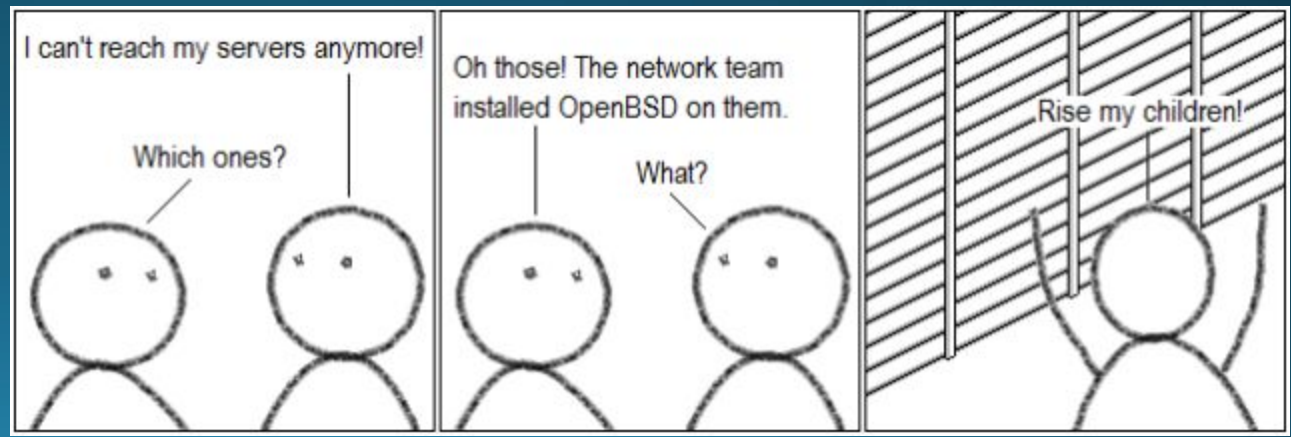
From DS Clear, To DS Clear:
Address 1: Destination
Address 2: Source
Address 3: BSSID

From DS Clear, To DS Set:
Address 1: BSSID
Address 2: Source
Address 3: Destination

From DS Set, To DS Set:
Address 1: Receiver
Address 2: Transmitter
Address 3: Destination
Address 4: Source

Crazy Memory Time

- Remember To First
  - 'Tis better TO give than to receive
- Next remember BSD
  - That crazy operating system TO give to your friends
    - BSSID
    - SOURCE
    - DESTINATION
      - To-DS



Crazy Memory Time

- Smart people use Databases (DBS), not Excel
  - Destination
  - BSSID
  - Source
    - From-DS

Crazy Memory Time

- Learn your Scapy Decimals for Direction
  - 1 - To-DS
  - 2 - From-DS
  - 65 - To-DS (with Encryption)
  - 66 - From-DS (with Encryption)

Crazy Memory Time

- Learn your Scapy Decimals for Type
  - 0 - Management
  - 1 - Control
  - 2 - Data

"First you have to Manage, then you can Control the Data"

Crazy Memory Time

- All of this corresponds to the Dot11 Class
  - FCfield
  - addr1
  - addr2
  - addr3
  - type
- Learn how to manipulate and interpret the above
  - The world becomes your oyster

TCP

- What do you think of when you hear TCP-IP?

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  - syn , syn/ack, ack
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- If you remember nothing else

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  - Sequence and Acknowledgment numbers
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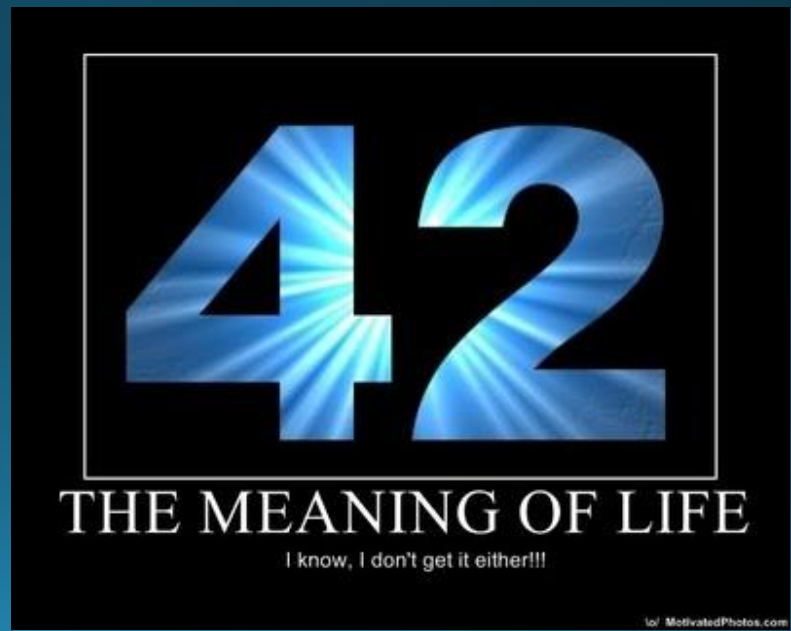
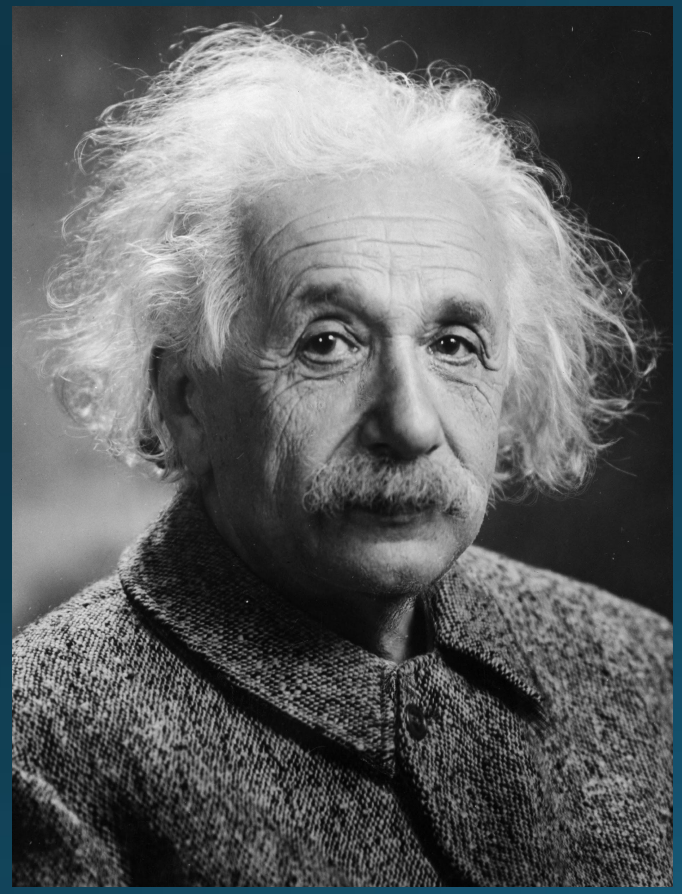
- If you remember nothing else
  - Sequence + Length == Acknowledgment
    - This is your 42



TCP Flags

- FIN – No more data from sender (Seen after a connection is closed)
- URG – indicates that the URGent pointer field is significant
- CWR – Congestion Window Reduced (CWR) flag is set by the sending host to indicate that it received a TCP segment with the ECE flag set (added to header by RFC 3168).
- ACK – indicates that the ACKnowledgment field is significant (Sometimes abbreviated by tcpdump as ".")
- SYN – Synchronize sequence numbers (Seen on new connections)
- PSH – Push function
- ECE (ECN-Echo) – indicate that the TCP peer is ECN capable during 3-way handshake (added to header by RFC 3168).
- RST – Reset the connection (Seen on rejected connections)

TCP #s



-VS-

ABSOLUT
Country of Sweden
VODKA
Every drop of this
superb vodka has been crafted
only with Swedish winter wheat
near the small town of Åhus
and continues a determined
commitment to the pursuit of
perfection since 1879.
IMPORTED
40% ALC./VOL. (80 PROOF) 700 ML
PRODUCED AND BOTTLED IN ÅHUS, SWEDEN

TCP #s

1	0.000000	192.168.1.125	192.168.1.132	TCP	74	37310 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM=1 TSval=37983574 TSecr=0 WS=1024
2	0.000396	192.168.1.132	192.168.1.125	TCP	74	80 → 37310 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM=1 TSval=359152 TSecr=37983574 WS=128
3	0.000452	192.168.1.125	192.168.1.132	TCP	66	37310 → 80 [ACK] Seq=1 Ack=1 Win=29696 Len=0 TSval=37983574 TSecr=359152
4	0.000544	192.168.1.125	192.168.1.132	HTTP	143	GET / HTTP/1.1
5	0.000790	192.168.1.132	192.168.1.125	TCP	66	80 → 37310 [ACK] Seq=1 Ack=78 Win=29056 Len=0 TSval=359152 TSecr=37983574
6	0.002040	192.168.1.132	192.168.1.125	HTTP	338	HTTP/1.1 200 OK (text/html)
7	0.002066	192.168.1.125	192.168.1.132	TCP	66	37310 → 80 [ACK] Seq=78 Ack=273 Win=30720 Len=0 TSval=37983574 TSecr=359152
8	0.002241	192.168.1.125	192.168.1.132	TCP	66	37310 → 80 [FIN, ACK] Seq=78 Ack=273 Win=30720 Len=0 TSval=37983574 TSecr=359152
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-VS-

```
Out[9]: <Ether  dst=aa:bb:cc:dd:ee:ff src=08:00:27:0d:60:76 type=0x800 |<IP  version=4L ihl=5L tos=0x0 len=52 id=50023 flags=DF frag=0L ttl=64 proto=t
cp chksum=0xf30a src=192.168.1.132 dst=192.168.1.125 options=[] |<TCP  sport=http dport=37310 seq=840022423 ack=454279889 dataofs=8L reserved=0L flags
=A window=227 chksum=0x835d urgptr=0 options=[('NOP', None), ('NOP', None), ('Timestamp', (359152, 37983574))] |>>>
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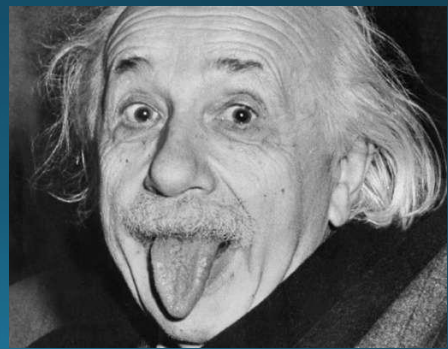

Scapy

- Created by Philippe Biondi
- Best interactive packet manipulator, ever.
- Who here has used?

Scapy tidbits

- Namespace overwrite warning…
  - ls()
- lsc()
- pkt.show()/pkt.show2()
- rdpcap()/wrpcap()
- PcapReader()/PcapWriter()
- hexstr(), binascii.un/hexlify()
- send()/sendp()
  - prn()
  - lfilter()

Learning to fly

- sendp() and send() only "work" if the tgt receives a frame which it expects
  - Garbage in, Garbage out
- So, how do we build?
  - We could learn by reading the RFC and building the packet from scratch…
  - OR:
  - We use a pre-existing packet/frame from a PCAP!
    - https://wiki.wireshark.org/samplecaptures
- Trust, but verify; test and often, as much as you can.

This is my Rifle

- We now know how to listen
- We wait for the trigger, then pull

airpwn-ng

- Conceptualized from the original airpwn by toast
- POC in bash
  - Python for the win!
- Why does it work?
  - 1st to the finish line of course…

pyDot11

- On-the-fly encryption and decryption of 802.11
  - WEP or WPA
    - CCMP only right now, no TKIP…
- Created because I could not find any other tool like it
  - airventriloquist-ng…
- Learned so much
- Planned deprecation in favor of airventriloquist-ng as a .so for Python

airtun-ng

- Because of course it existed

Tools

- edgeDressing
- foxHunter
- frameTracer
- packetEssentials
- pmkid2hashcat
- piCopilot